

Bulk DWBA Scattering and the Specular Mean-Density Correction

orGUI design note

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Abstract

This note derives the first-order distorted-wave Born approximation (DWBA) used by orGUI from the complex X-ray refractive index through the far-field electric field. It fixes the dimensional prefactors in the relation between atomic scattering factors and the refractive index, states a self-consistent Green-function sign convention, and identifies the quantity currently returned by `F_DWBA`. For a semi-infinite bulk, the atomic density must be measured relative to the homogeneous density already used to generate the reference Fresnel field. On the specular rod this produces an additional term proportional to $1/(iQ_z)$. The term vanishes off-specular because a laterally homogeneous reference has no Fourier component at nonzero in-plane momentum transfer.

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1 Scope and conventions

The purpose of this note is to establish the normalization needed before the bulk DWBA structure factor can be combined coherently with the reference specular reflection amplitude. The derivation concerns a nonmagnetic, isotropic, linear, reciprocal medium. Magnetic and tensor-resonant scattering require a corresponding tensor susceptibility and are outside the present scope.

We use the crystallographic space-time convention

$$\mathbf{E}(\mathbf{r}, t) \propto \exp[i(\omega t - \mathbf{k} \cdot \mathbf{r})], \quad (1)$$

with

$$k_0 = \frac{2\pi}{\lambda}, \quad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i. \quad (2)$$

The Fourier transform used for a density $g(\mathbf{r})$ is

$$\tilde{g}(\mathbf{Q}) = \int g(\mathbf{r}) \exp(i\mathbf{Q} \cdot \mathbf{r}) \, d^3r. \quad (3)$$

The surface normal is the z direction. Vacuum lies at $z > 0$, and the bulk extends toward negative z . The normal wavevectors in orGUI follow the Renaud sign convention

$$k_{z,j} = -k_0 \sqrt{n_j^2 - n_0^2 \cos^2 \alpha}, \quad \operatorname{Re} k_{z,j} \leq 0, \quad \operatorname{Im} k_{z,j} \geq 0. \quad (4)$$

The branch choice makes waves propagating or decaying into the sample behave correctly with Eq. (1).

All electric-field amplitudes in the following equations are normalized to a unit incident plane-wave amplitude unless stated otherwise. A selected incident and analyzed exit polarization are treated explicitly. The scalar formulae are recovered by replacing the vector contraction by ordinary multiplication.

2 Refractive index and scattering-factor density

2.1 Atomic scattering factors

For atom j , the complex X-ray scattering factor is written

$$f_j(\mathbf{Q}, \lambda) = f_j^0(\mathbf{Q}) + f_j'(\lambda) + i f_j''(\lambda). \quad (5)$$

Here f_j^0 is the normal, momentum-dependent atomic form factor, while f_j' and f_j'' are the energy-dependent anomalous dispersion terms. The quantity $f_j^0(\mathbf{0})$ equals the number of electrons of the neutral atom when ionic corrections are absent.

It is useful to define a complex scattering-factor density

$$\rho_f(\mathbf{r}) = \sum_j o_j \mathcal{F}^{-1}[f_j(\mathbf{Q}, \lambda)](\mathbf{r} - \mathbf{r}_j), \quad (6)$$

where o_j is the occupancy and $\mathbf{r}_j$ is the atomic position. The units of ρ_f are electrons per cubic Angstrom when the scattering factors are expressed in electron units. Thermal disorder is applied in reciprocal space through the Debye–Waller factor and therefore need not be included in the real-space definition in Eq. (6).

2.2 Correct refractive-index prefactor

Let r_e denote the classical electron radius,

$$r_e = 2.8179403262 \times 10^{-5} \text{ Angstrom.} \quad (7)$$

For the convention in Eq. (1), the refractive index is

$$n(\mathbf{r}) = 1 - \frac{2\pi r_e}{k_0^2} \rho_f(\mathbf{r}). \quad (8)$$

For a homogeneous atomic number density ρ_a , this becomes

$$n = 1 - \delta - i\beta = 1 - \frac{2\pi r_e \rho_a}{k_0^2} [f^0(0) + f' + if'']. \quad (9)$$

The factor $2\pi r_e \rho_a / k_0^2$ is dimensionless. A coefficient proportional to $r_e \rho_a / k_0$ would have dimensions of inverse length and cannot be a refractive-index decrement.

Because $|n - 1| \ll 1$ for hard X-rays, terms quadratic in the susceptibility may be neglected:

$$n^2(\mathbf{r}) \simeq 1 - \frac{4\pi r_e}{k_0^2} \rho_f(\mathbf{r}). \quad (10)$$

2.3 Homogeneous bulk reference and perturbation

For the simple bulk problem, the unperturbed reference medium is vacuum above a homogeneous semi-infinite substrate:

$$n_0(z) = \begin{cases} 1, & z > 0, \\ 1 - \delta_0 - i\beta_0, & z < 0. \end{cases} \quad (11)$$

The equivalent homogeneous complex scattering-factor density is

$$\bar{\rho}_f = \frac{k_0^2}{2\pi r_e} (\delta_0 + i\beta_0). \quad (12)$$

For a bulk unit cell of volume V_{uc} , it can also be calculated as

$$\bar{\rho}_f = \frac{1}{V_{uc}} \sum_{j \in uc} o_j [f_j^0(0) + f_j' + if_j'']. \quad (13)$$

Let $\Theta(-z)$ be one inside the substrate and zero in vacuum. Splitting the actual index into reference and perturbation parts,

$$n^2(\mathbf{r}) = n_0^2(z) + \Delta n^2(\mathbf{r}), \quad (14)$$

gives the corrected perturbation

$$\begin{aligned} \Delta n^2(\mathbf{r}) &= n^2(\mathbf{r}) - n_0^2(z) \\ &= -\frac{4\pi r_e}{k_0^2} [\rho_f(\mathbf{r}) - \bar{\rho}_f \Theta(-z)]. \end{aligned} \quad (15)$$

Equivalently,

$$\Delta n^2(\mathbf{r}) = -\frac{4\pi r_e}{k_0^2} \rho_f(\mathbf{r}) + (2\delta_0 + 2i\beta_0)\Theta(-z). \quad (16)$$

The reference term is present only in the material half-space. Treating it as a constant throughout all space would incorrectly reduce it to a delta function only at $\mathbf{Q} = 0$ and would erase the specular surface contribution.

3 Helmholtz equation and the Green function

3.1 Reference Helmholtz operator

For a nonmagnetic medium, the electric field obeys

$$[\nabla^2 + k_0^2 n^2(\mathbf{r})] \mathbf{E}(\mathbf{r}) = 0. \quad (17)$$

Define the reference operator

$$\mathcal{L}_0 = \nabla^2 + k_0^2 n_0^2(\mathbf{r}). \quad (18)$$

Using Eq. (15), the exact field satisfies

$$\mathcal{L}_0 \mathbf{E}(\mathbf{r}) = -k_0^2 \Delta n^2(\mathbf{r}) \mathbf{E}(\mathbf{r}). \quad (19)$$

3.2 Green-function normalization and sign

We define the outgoing dyadic Green function by

$$\boxed{\mathcal{L}_0 \mathbf{G}_0(\mathbf{r}, \mathbf{r}') = -\frac{k_0^2}{\epsilon_0} \mathbf{I} \delta(\mathbf{r} - \mathbf{r}').} \quad (20)$$

Here ϵ_0 is the vacuum permittivity and $\mathbf{I}$ is the identity tensor. The negative source sign is intentional. With the Fourier and time conventions above, it makes Eq. (20) consistent with both Eq. (19) and the positive kernel in the Lippmann–Schwinger equation below.

Other references sometimes define the Green function with the opposite point source or define $\Delta n^2 = n_0^2 - n^2$. Either choice is permissible, but all subsequent signs must then change consistently. The phase of the final specular correction must be checked against a Fresnel limiting case because it controls interference with the reference reflection amplitude.

3.3 Far-field reciprocal form

Let the observation point be in vacuum at distance r in the outgoing direction $\mathbf{k}_f$. For an analyzed exit polarization ϵ_f , reciprocity gives the selected-polarization action of the Green tensor as

$$\mathbf{G}_0(\mathbf{r}, \mathbf{r}') \mathbf{p} \xrightarrow{r \rightarrow \infty} \frac{k_0^2 e^{-ik_0 r}}{4\pi \epsilon_0 r} \epsilon_f [\mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f) \cdot \mathbf{p}]. \quad (21)$$

The field $\mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f)$ is the reciprocal exit field generated by a plane wave incident from the detector direction with wavevector $-\mathbf{k}_f$. It is calculated in the same complex reference profile as the incident field. It is not complex conjugated; absorption does not invalidate the reciprocity construction for the reciprocal, linear medium considered here.

4 Lippmann–Schwinger equation and Born series

Let $\mathbf{E}_0(\mathbf{r}, \mathbf{k}_i)$ be the exact solution in the reference medium for the incident wavevector $\mathbf{k}_i$. Applying the Green function to Eq. (19) gives the exact integral equation

$$\boxed{\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\mathbf{r}) + \epsilon_0 \int \mathbf{G}_0(\mathbf{r}, \mathbf{r}') \Delta n^2(\mathbf{r}') \mathbf{E}(\mathbf{r}') d^3 r'.} \quad (22)$$

Iterating Eq. (22) produces the Born series:

$$\begin{aligned}
\mathbf{E}(\mathbf{r}) = & \mathbf{E}_0(\mathbf{r}) \\
& + \epsilon_0 \int \mathbf{G}_0(\mathbf{r}, \mathbf{r}') \Delta n^2(\mathbf{r}') \mathbf{E}_0(\mathbf{r}') d^3 r' \\
& + \epsilon_0^2 \iint \mathbf{G}_0(\mathbf{r}, \mathbf{r}') \Delta n^2(\mathbf{r}') \mathbf{G}_0(\mathbf{r}', \mathbf{r}'') \Delta n^2(\mathbf{r}'') \mathbf{E}_0(\mathbf{r}'') d^3 r' d^3 r'' \\
& + \dots
\end{aligned} \tag{23}$$

Each additional term adds another scattering event from the perturbation. Reflection, transmission, refraction, and repeated reflection at the planar interfaces are already included to all orders in $\mathbf{E}_0$ and $\mathbf{G}_0$. DWBA is “Born” only with respect to the residual contrast Δn^2 .

First-order DWBA retains the first correction:

$$\delta \mathbf{E}^{(1)}(\mathbf{r}) = \epsilon_0 \int \mathbf{G}_0(\mathbf{r}, \mathbf{r}') \Delta n^2(\mathbf{r}') \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) d^3 r'. \tag{24}$$

Substituting the far-field Green function from Eq. (21) yields

$$\begin{aligned}
\delta \mathbf{E}_f(\mathbf{r}) = & \frac{k_0^2 e^{-ik_0 r}}{4\pi r} \boldsymbol{\epsilon}_f \\
& \times \int \mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f) \cdot \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) \Delta n^2(\mathbf{r}') d^3 r'.
\end{aligned} \tag{25}$$

Equation (25) is the field-level DWBA master equation. Its squared matrix element gives the usual differential scattering cross-section, but the equation itself retains the phase needed for coherent addition to another outgoing field.

5 Insertion of the refractive-index perturbation

Inserting Eq. (15) into Eq. (25) cancels the factor $k_0^2/(4\pi)$:

$$\boxed{\delta \mathbf{E}_f(\mathbf{r}) = -r_e \frac{e^{-ik_0 r}}{r} \boldsymbol{\epsilon}_f F_{\Delta, \text{DWBA}}.} \tag{26}$$

The contrast DWBA amplitude in electron units is

$$\begin{aligned}
F_{\Delta, \text{DWBA}} = & \int \mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f) \cdot \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) \\
& \times [\rho_f(\mathbf{r}') - \bar{\rho}_f \Theta(-z')] d^3 r'.
\end{aligned} \tag{27}$$

It is useful to split this into two terms:

$$F_{\Delta, \text{DWBA}} = F_{\text{atomic}, \text{DWBA}} - F_{\text{mean}, \text{DWBA}}, \tag{28}$$

where

$$F_{\text{atomic}, \text{DWBA}} = \int \mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f) \cdot \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) \rho_f(\mathbf{r}') d^3 r', \tag{29}$$

$$F_{\text{mean}, \text{DWBA}} = \bar{\rho}_f \int_{z' < 0} \mathbf{E}_0(\mathbf{r}', -\mathbf{k}_f) \cdot \mathbf{E}_0(\mathbf{r}', \mathbf{k}_i) d^3 r'. \tag{30}$$

For a finite scatterer and a unit incident field, the corresponding differential cross-section is

$$\frac{d\sigma}{d\Omega} = r_e^2 |F_{\Delta, \text{DWBA}}|^2. \tag{31}$$

The local-field amplitudes and polarization contraction are already contained in $F_{\Delta, \text{DWBA}}$. Equation (31) shows why an additional factor $k_0^2/(16\pi^2)$ must not remain after converting the dielectric perturbation into atomic electron units.

6 Four Renaud channels and the current orGUI amplitude

In an optical medium j , both the incident and reciprocal exit fields contain downward and upward components. Let

$$\sigma_i, \sigma_f \in \{+1, -1\} \quad (32)$$

label those components. orGUI orders the four channels as

$$++, \quad +-, \quad -+, \quad --. \quad (33)$$

For channel $\nu = (\sigma_i, \sigma_f)$, the normal scattering transfer is

$$Q_{z,j}^\nu = -(\sigma_i k_{z,j,i} + \sigma_f k_{z,j,f}). \quad (34)$$

The channel coefficient is schematically

$$\begin{aligned} C_{j,\nu} &= A_{j,i}^{\sigma_i} A_{j,f}^{\sigma_f} \left(\epsilon_{j,f}^{\sigma_f} \cdot \epsilon_{j,i}^{\sigma_i} \right) \\ &\times \exp \left[i \left(\sigma_i k_{z,j,i} z_{j,i}^{\text{ref}} + \sigma_f k_{z,j,f} z_{j,f}^{\text{ref}} \right) \right]. \end{aligned} \quad (35)$$

The reference-position phase in Eq. (35) is required because the stored amplitudes are local amplitudes referenced to z_j^{ref} rather than to a global origin.

The atomic contribution can then be written

$$F_{\text{atomic,DWBA}} = \sum_{\nu} \sum_a C_{j(a),\nu} o_a f_a (Q_{j(a),\nu}^2) e^{-W_a(Q_{j(a),\nu}^2)} e^{i \mathbf{Q}_{j(a),\nu} \cdot \mathbf{r}_a}, \quad (36)$$

including the coherent-domain sum when more than one domain is present. Every channel generally has a different complex Q_z and therefore a different complex Q^2 .

The quantity currently returned by $\mathbf{F_DWBA}$ is $F_{\text{atomic,DWBA}}$, scaled to electrons per configured reference lateral unit cell. It includes the electric-field amplitudes, polarization, atomic form factors, Debye–Waller factors, atomic coordinates, coherent domains, and the bulk lattice sum. It does not yet subtract $F_{\text{mean,DWBA}}$.

For an atom in the terminal substrate,

$$A_i^- = A_f^- = 0. \quad (37)$$

Only the $++$ channel survives, and its coefficient will be denoted simply by $\mathcal{C}_{++}$.

7 Semi-infinite bulk and the new $1/(iQ_z)$ term

7.1 Relation to the dynamical-theory χ_0 term

The subtraction derived in this section is not a bookkeeping convention. It is the DWBA restatement of a step that dynamical diffraction theory performs at the very start, and the argument is clearest in that setting.

Dynamical theory expands the triply periodic susceptibility $\chi = n^2 - 1$ over reciprocal lattice vectors $\mathbf{H}$,

$$\chi(\mathbf{r}) = \sum_{\mathbf{H}} \chi_{\mathbf{H}} e^{i\mathbf{H} \cdot \mathbf{r}}, \quad \chi_{\mathbf{H}} = -\frac{4\pi r_e}{k_0^2 V_{\text{uc}}} F_{\mathbf{H}}, \quad (38)$$

so that by Eq. (10) the $\mathbf{H} = \mathbf{0}$ coefficient is precisely the mean density of Eq. (13),

$$\chi_0 = -\frac{4\pi r_e}{k_0^2} \bar{\rho}_f = -2(\delta_0 + i\beta_0). \quad (39)$$

Inserting the Bloch ansatz $\mathbf{E} = \sum_{\mathbf{H}} \mathbf{E}_{\mathbf{H}} \exp(-i\mathbf{k}_{\mathbf{H}} \cdot \mathbf{r})$ into Eq. (17) gives the fundamental equations of dynamical theory,

$$(k_{\mathbf{H}}^2 - k_0^2) \mathbf{E}_{\mathbf{H}} = k_0^2 \sum_{\mathbf{H}'} \chi_{\mathbf{H}-\mathbf{H}'} \mathbf{E}_{\mathbf{H}'}. \quad (40)$$

The term $\mathbf{H}' = \mathbf{H}$ on the right-hand side carries the single coefficient χ_0 and couples no two distinct Fourier components. It is therefore moved to the left and absorbed into the propagation constant,

$$K^2 = k_0^2 (1 + \chi_0), \quad (k_{\mathbf{H}}^2 - K^2) \mathbf{E}_{\mathbf{H}} = k_0^2 \sum_{\mathbf{H}' \neq \mathbf{H}} \chi_{\mathbf{H}-\mathbf{H}'} \mathbf{E}_{\mathbf{H}'}. \quad (41)$$

Equation (41) contains the entire argument. In dynamical theory χ_0 is a redefinition of the dispersion relation, not a source term. Refraction, the critical angle, absorption, and the Fresnel or Parratt specular amplitude are all consequences of its left-hand side. Only the coefficients with $\mathbf{H} \neq \mathbf{0}$ ever act as scattering.

The specular rod is the unique rod that contains the point $\mathbf{H} = \mathbf{0}$. Every other rod passes only through reciprocal lattice points with $\mathbf{H}_{\parallel} \neq \mathbf{0}$. That is the structural reason for the two consequences drawn from Eq. (50) below, and it explains why the published DWBA treatments of crystal truncation rods, which work on non-specular rods, never need the subtraction.

7.2 The exact Fresnel amplitude in first-Born form

The one-beam limit of Eq. (41), in which the sum on the right is empty, is solved exactly by matching tangential fields at the surface. For a single interface the result rearranges into

$$\begin{aligned} r_0 &= \frac{k_{z,0} - k_{z,1}}{k_{z,0} + k_{z,1}} = \frac{k_{z,0}^2 - k_{z,1}^2}{(k_{z,0} + k_{z,1})^2} \\ &= \frac{-k_0^2 \chi_0}{(k_{z,0} + k_{z,1})^2} = \frac{4\pi r_e \bar{\rho}_f}{(k_{z,0} + k_{z,1})^2}, \end{aligned} \quad (42)$$

where $k_{z,1}^2 = k_{z,0}^2 + k_0^2 \chi_0$ was used. Equation (42) is exact, not perturbative. It states that the exact Fresnel amplitude is the first-Born answer with the vacuum Q_z replaced by the sum of the in-medium normal wavevectors. For $k_{z,1} \rightarrow k_{z,0} = \kappa_0$ the denominator becomes Q_z^2 and

$$r_0 \longrightarrow \frac{4\pi r_e \bar{\rho}_f}{Q_z^2} = \frac{\pi r_e \bar{\rho}_f}{\kappa_0^2}, \quad (43)$$

which is the limit used as a phase check in Eq. (67).

7.3 The mean-density term as the $\mathbf{H} = \mathbf{0}$ pole of the truncation sum

The same term can be exhibited directly inside the kinematical lattice sum, which makes the $1/(iQ_z)$ factor a derived object rather than a distributional identity. The semi-infinite stacking sum used in Eq. (52) below,

$$S(Q_z) = \sum_{m=0}^{\infty} e^{-iQ_z m c} = \frac{1}{1 - e^{-iQ_z c}}, \quad (44)$$

is a meromorphic function of the continuous rod coordinate $L = Q_z c / (2\pi)$ with a simple pole at every integer l , that is, at every reciprocal lattice point along the rod. Differentiating the denominator gives the residue of $F_{hkl} S$ at $L = l$ as

$$\text{Res}_{L=l} \frac{F_{hkl}}{1 - e^{-2\pi i L}} = \frac{F_{hkl}}{2\pi i}. \quad (45)$$

Kaganer shows that the corresponding pole expansion is exact, and that each pole term is the asymptotic two-beam dynamical amplitude of the reflection hkl far from its Bragg peak. The kinematical crystal truncation rod is therefore a coherent sum of dynamical Bragg tails.

On a non-specular rod every pole sits at a reciprocal lattice point with $\mathbf{H}_{\parallel} \neq \mathbf{0}$. On the specular rod one pole sits at $l = 0$, and its residue is the total scattering power of the unit cell,

$$F_{000} = \sum_{a \in \text{uc}} o_a f_a(0) = V_{\text{uc}} \bar{\rho}_f = A_{\text{ref}} c \bar{\rho}_f. \quad (46)$$

Isolating that single pole, with $2\pi L = Q_z c$, gives

$$\frac{F_{000}}{2\pi i} \frac{1}{L} = \frac{F_{000}}{i Q_z c} = \frac{A_{\text{ref}} \bar{\rho}_f}{i Q_z}, \quad (47)$$

which is exactly the mean-density factor obtained independently in Eq. (54) below. Carrying Eq. (47) through the plane-wave normalization of Eq. (61) yields

$$\delta r_{\text{mean}} = \frac{2\pi i r_e}{\kappa_0 A_{\text{ref}}} \frac{A_{\text{ref}} \bar{\rho}_f}{i Q_z} = \frac{2\pi r_e \bar{\rho}_f}{\kappa_0 Q_z} = \frac{4\pi r_e \bar{\rho}_f}{Q_z^2}, \quad (48)$$

identical to the high-angle Fresnel amplitude of Eq. (43).

The relation is therefore an identity, not an analogy. The $1/(i Q_z)$ term is the $\mathbf{H} = \mathbf{0}$ ‘‘Bragg reflection’’ of the crystal, and the $\mathbf{H} = \mathbf{0}$ Bragg reflection is specular reflection. Dynamical theory disposes of it by absorbing χ_0 into the refractive index, as in Eq. (41). Kinematical theory disposes of it by leaving it inside the lattice sum. Exactly one of the two must be done. Because r_0 in Eq. (68) is taken from the exact layered solution, the first route has already been chosen, and the term must be removed from the second.

Two practical consequences follow. First, well above the critical angle and away from Bragg peaks $\delta r_{\text{mean}} \simeq r_0$, so an unsubtracted specular rod returns $r_{\text{total}} \simeq 2r_0$ and a reflectivity too large by roughly a factor of four. This is a sharper diagnostic than visual inspection of a fit. Second, as $Q_z \rightarrow 0$ the kinematical specular sum diverges like $A_{\text{ref}} \bar{\rho}_f / (i Q_z)$, whereas the true reflectivity saturates at $R = 1$ below the critical angle. The kinematical specular rod is thus unphysical at small Q_z irrespective of any surface model. Removing the divergent pole and replacing it by the bounded exact r_0 is what repairs this.

7.4 Why a truncated constant is not only a point at $Q = 0$

For a constant extending over all three-dimensional space,

$$\int_{\mathbb{R}^3} C e^{i\mathbf{Q} \cdot \mathbf{r}} d^3 r = (2\pi)^3 C \delta^{(3)}(\mathbf{Q}). \quad (49)$$

The reference density, however, exists only for $z < 0$. Its transform factorizes as

$$\begin{aligned} \int d^2 r_{\parallel} e^{i\mathbf{Q}_{\parallel} \cdot \mathbf{r}_{\parallel}} \int_{-\infty}^0 dz C e^{iQ_z z} \\ = (2\pi)^2 \delta^{(2)}(\mathbf{Q}_{\parallel}) \frac{C}{i Q_z}, \end{aligned} \quad (50)$$

where the complex DWBA Q_z supplies physical convergence through absorption. For a real Q_z , Eq. (50) is understood with the usual infinitesimal convergence factor and corresponding distributional limit.

Equation (50) has two immediate consequences:

1. For $\mathbf{Q}_{\parallel} \neq \mathbf{0}$, the laterally homogeneous reference term is zero. The existing atomic F_{DWBA} is therefore already the contrast amplitude for a non-specular CTR.
2. For $\mathbf{Q}_{\parallel} = \mathbf{0}$, the reference contributes along the complete specular rod. It cannot be discarded as a term only at $Q_z = 0$.

7.5 Unit-cell and lattice-sum derivation

Choose the top bulk unit cell to occupy $-c \leq z < 0$, where $c > 0$ is the surface-normal repeat length, and let A_{ref} be its configured reference lateral area. Define the atomic unit-cell factor, before applying the $++$ field coefficient, by

$$F_{\text{uc}}^{\text{atomic}}(Q_z) = \sum_{a \in \text{uc}} o_a f_a(Q^2) e^{-W_a(Q^2)} e^{i\mathbf{Q} \cdot \mathbf{r}_a}. \quad (51)$$

The atomic semi-infinite bulk amplitude is

$$F_{\text{atomic,DWBA}} = \mathcal{C}_{++} \frac{F_{\text{uc}}^{\text{atomic}}(Q_z)}{1 - e^{-iQ_z c}}. \quad (52)$$

The mean-density contribution from the same top unit cell is

$$\begin{aligned} F_{\text{mean,uc}}(Q_z) &= A_{\text{ref}} \bar{\rho}_f \int_{-c}^0 e^{iQ_z z} dz \\ &= A_{\text{ref}} \bar{\rho}_f \frac{1 - e^{-iQ_z c}}{iQ_z}. \end{aligned} \quad (53)$$

Applying the same semi-infinite lattice sum gives

$$\begin{aligned} F_{\text{mean,DWBA}} &= \mathcal{C}_{++} \frac{F_{\text{mean,uc}}(Q_z)}{1 - e^{-iQ_z c}} \\ &= \boxed{\mathcal{C}_{++} \frac{A_{\text{ref}} \bar{\rho}_f}{iQ_z}}. \end{aligned} \quad (54)$$

The cancellation between numerator and denominator is the unit-cell version of the half-space transform in Eq. (50).

Combining Eqs. (52) and (54) gives the required contrast amplitude:

$$F_{\Delta, \text{DWBA}} = \mathcal{C}_{++} \left[\frac{F_{\text{uc}}^{\text{atomic}}(Q_z)}{1 - e^{-iQ_z c}} - \frac{A_{\text{ref}} \bar{\rho}_f}{iQ_z} \right], \quad \mathbf{Q}_{\parallel} = \mathbf{0}. \quad (55)$$

An equivalent cellwise form is

$$F_{\Delta, \text{DWBA}} = \mathcal{C}_{++} \frac{F_{\text{uc}}^{\text{atomic}}(Q_z) - A_{\text{ref}} \bar{\rho}_f \frac{1 - e^{-iQ_z c}}{iQ_z}}{1 - e^{-iQ_z c}}. \quad (56)$$

The cellwise form makes the cancellation of the cell-average density explicit. As $Q_z \rightarrow 0$,

$$\frac{1 - e^{-iQ_z c}}{iQ_z} \rightarrow c, \quad (57)$$

and

$$F_{\text{uc}}^{\text{atomic}}(0) = A_{\text{ref}} c \bar{\rho}_f. \quad (58)$$

Thus the leading mean-density contribution cancels before the lattice sum is formed. Numerically, both the lattice denominator and the finite-cell mean factor should be evaluated with complex `expm1`-type expressions to avoid cancellation error near $Q_z c = 0$.

The relation to the current bulk API is therefore

$$F_{\Delta, \text{DWBA}} = F_{\text{DWBA}}^{\text{current}} - \mathcal{C}_{++} \frac{A_{\text{ref}} \bar{\rho}_f}{iQ_z}, \quad \mathbf{Q}_{\parallel} = \mathbf{0}. \quad (59)$$

7.6 Truncation of the atomic densities at the surface plane

Equation (15) subtracts a reference density that is sharply truncated at $z = 0$, whereas $\rho_f(\mathbf{r})$ is a superposition of smooth atomic densities. The cancellation in Eq. (56) is therefore exact only if the electron density of the top unit cell is contained entirely in $z < 0$. If the surface plane bisects an atom, a residual contrast of the order of the truncated electron fraction of that atom survives at all Q_z , because the atomic term retains the part of the cloud at $z > 0$ while the reference term does not.

Kaganer reports the same effect when comparing kinematical and dynamical CTR calculations: the residual limits the achievable agreement to roughly 10^{-7} of the peak intensity, and it is removed by choosing the cell origin so that the mathematical surface falls between atomic layers rather than through one. For GaAs(001) this means placing the atoms at fractional depths $1/8, 3/8, 5/8, 7/8$ rather than $0, 1/4, 1/2, 3/4$.

The consequence here is that the bulk cell used to generate $\bar{\rho}_f$ and the bulk cell used to generate $F_{\text{uc}}^{\text{atomic}}$ must share one origin, chosen to minimise atomic truncation at $z = 0$. Invariant 8 below states the corresponding test.

8 Observable specular field

Equation (26) describes spherical far-field radiation from a finite coherent scattering volume. An ideal laterally infinite specular interface instead produces a plane reflected wave. The lateral integral supplies $\delta^{(2)}(\mathbf{Q}_{\parallel})$, and the result must be normalized as a dimensionless reflection coefficient rather than by $e^{-ik_0 r}/r$.

Let

$$\kappa_0 = k_0 \sin \alpha = -k_{z,0} > 0 \quad (60)$$

be the positive magnitude of the vacuum normal wavevector. For the Fourier phase in Eq. (3), the plane-wave correction per reference lateral cell is

$$\delta r_{\text{DWBA}} = \frac{2\pi i r_e}{\kappa_0} \frac{F_{\Delta, \text{DWBA}}}{A_{\text{ref}}}. \quad (61)$$

The prefactor has the required dimensions:

$$\frac{r_e}{\kappa_0} \frac{F}{A} \sim \frac{\text{Angstrom}}{\text{Angstrom}^{-1}} \frac{\text{electrons}}{\text{Angstrom}^2}, \quad (62)$$

which is dimensionless when electron units are understood for F .

8.1 Comparison with the published kinematical prefactor

The prefactor in Eq. (61) can be checked against an independently derived semi-infinite kinematical amplitude. Kaganer, in the work listed in Section 12, gives the plane-wave amplitude per lateral cell as

$$E_{\text{kin}}(hkL) = \frac{r_e \lambda}{a^2 \gamma_f} \frac{i F_{hkL}}{\exp(2\pi i L) - 1}, \quad (63)$$

where a^2 is the lateral cell area, $\gamma_f = \sin \alpha_f$ is the sine of the exit angle, and $2\pi L = Q_z c$. Substituting $\lambda = 2\pi/k_0$ and $\kappa_f = k_0 \gamma_f$ gives

$$\frac{r_e \lambda}{a^2 \gamma_f} = \frac{2\pi r_e}{\kappa_f A_{\text{ref}}}, \quad (64)$$

so that Eq. (63) becomes

$$E_{\text{kin}} = \frac{2\pi i r_e}{\kappa_f} \frac{1}{A_{\text{ref}}} \frac{F_{hkL}}{\exp(i Q_z c) - 1}. \quad (65)$$

This is Eq. (61) with the atomic lattice sum inserted. The agreement is exact, including the factor i and the absence of any residual $k_0^2/(4\pi)$, and it corroborates invariant 1 below against a result that has itself been benchmarked against multi-beam dynamical diffraction theory.

Two conventions differ harmlessly but must be respected. First, the lattice denominator $\exp(iQ_z c) - 1$ and the denominator $1 - \exp(-iQ_z c)$ used here differ by one factor $\exp(-iQ_z c)$, that is, by a shift of the unit-cell origin by one repeat. Because δr_{DWBA} is added coherently to r_0 , this phase is not free: the origin used for $F_{\text{uc}}^{\text{atomic}}$, for the $\Theta(-z)$ step, and for r_0 must be one and the same. Second, the normalizing wavevector in Eq. (63) is the *exit* normal component κ_f . On the specular rod $\kappa_f = \kappa_0$ and Eq. (61) is correct as written, but the same normalization must not be reused off-specular without replacing κ_0 by κ_f .

As a phase check, take a vacuum reference and a homogeneous half-space of scattering-factor density ρ_f . Then

$$\frac{F}{A} = \frac{\rho_f}{iQ_z}, \quad Q_z = 2\kappa_0, \quad (66)$$

and Eq. (61) gives

$$\delta r = \frac{\pi r_e \rho_f}{\kappa_0^2}, \quad (67)$$

which is the high-angle Fresnel amplitude to first order, in agreement with Eq. (43). The check is asymptotic: by Eq. (42) it holds only for $k_{z,1} \rightarrow k_{z,0}$, that is for Q_z well above the critical wavevector, and it must not be evaluated near α_c . This limit should be retained as an executable regression test when the specular correction is implemented.

For the actual DWBA decomposition, the total reflected amplitude is

$$\boxed{r_{\text{total}}(\alpha) = r_0(\alpha) + \delta r_{\text{DWBA}}(\alpha)}, \quad (68)$$

where r_0 must come from the same prepared optical reference used to compute $F_{\Delta, \text{DWBA}}$. In the current architecture this is the complex amplitude `prepared.field_i.r_S`, not the square root of a separately calculated reflectivity.

The coherent observable reflectivity is

$$R_{\text{coh}} = |r_0 + \delta r_{\text{DWBA}}|^2. \quad (69)$$

A formally first-order expansion of the intensity would retain only

$$R^{(1)} = |r_0|^2 + 2 \text{Re} [r_0^* \delta r_{\text{DWBA}}]. \quad (70)$$

Using Eq. (69) additionally includes the $|\delta r_{\text{DWBA}}|^2$ term, which is second order in the perturbation. First-order DWBA does not in general enforce exact energy conservation. In addition, absorption permits $R < 1$ below the critical angle, and roughness can transfer energy from the coherent specular beam into diffuse scattering.

For unpolarized radiation, s and p intensities are averaged incoherently:

$$R_{\text{unpolarized}} = \frac{1}{2} (|r_s|^2 + |r_p|^2). \quad (71)$$

The s and p amplitudes must not be added coherently.

9 Absorption and empirical attenuation

The normal transfer Q_z is generally complex because $k_{z,i}$ and $k_{z,f}$ are calculated from complex refractive indices. For $z < 0$, the factor

$$e^{iQ_z z} \quad (72)$$

therefore already contains the exact incident-plus-exit attenuation described by the reference DWBA fields. A separate factor such as $\exp[z/\Lambda(\alpha_i, \alpha_f)]$ would double-count absorption if used at the same time.

An explicit empirical attenuation remains meaningful for a purely kinematical comparison in which Q_z is deliberately kept real. If used, it must be applied continuously both within the top unit cell and between bulk repeats. For the exact DWBA path it should default to zero.

10 Implementation interpretation and validation invariants

The mathematical separation suggests keeping three quantities distinct in the public and native APIs:

1. $F_{\text{atomic,DWBA}}$: the atomic matrix element currently returned by `F_DWBA`;
2. $F_{\Delta,\text{DWBA}}$: the atomic matrix element after subtraction of the optical reference density;
3. δr_{DWBA} : the dimensionless specular reflection correction after plane-wave normalization.

The total reflected amplitude and reflectivity should be assembled only after these distinctions are explicit.

The following invariants are suitable for regression tests:

1. **Dimensional prefactor.** Substituting $\Delta n^2 = -4\pi r_e \Delta \rho_f / k_0^2$ into the DWBA field must leave the Thomson factor r_e , not $r_e k_0 / (4\pi)$.
2. **Identical actual and reference media.** If the actual density equals the homogeneous reference density, then $F_{\Delta,\text{DWBA}} = 0$ and $r_{\text{total}} = r_0$.
3. **Off-specular invariance.** For $Q_{\parallel} \neq 0$, adding the mean-density subtraction must not change the existing CTR amplitude.
4. **Low- Q_z cancellation.** The atomic cell average and the continuum cell factor must cancel without catastrophic numerical loss.
5. **High-angle Fresnel limit.** A homogeneous half-space treated relative to vacuum must reproduce the first-order Fresnel reflection amplitude, including its complex phase.
6. **Prepared-reference consistency.** The r_0 added to δr_{DWBA} must be taken from the same prepared refractive profile and polarization state as the DWBA fields.
7. **No duplicate attenuation.** Complex Q_z and explicit empirical attenuation must not both describe the same absorption.
8. **Surface truncation of atomic densities.** Shifting the bulk cell origin so that the surface plane no longer bisects an atom must not change $F_{\Delta,\text{DWBA}}$ by more than the level set by the truncated electron fraction. The same origin must be used for $F_{\text{uc}}^{\text{atomic}}$, for $\bar{\rho}_f \Theta(-z)$, and for the reference amplitude r_0 .

11 Symbol glossary

Symbol	Meaning
A_{ref}	Configured reference lateral unit-cell area, in square Angstrom.

Symbol	Meaning
$A_i^\pm, A_f^\pm$	Downward (+) and upward (−) local electric-field amplitudes for the incident and reciprocal exit solutions.
α	External glancing angle measured from the sample surface, in radians.
c	Positive surface-normal bulk repeat length. Repeated cells extend by $-c$ into the substrate.
$\mathcal{C}_{j,\nu}$	Complete field, polarization, and reference-phase coefficient for medium j and channel ν .
δ, β	Positive real refractive-index decrement and absorption index in $n = 1 - \delta - i\beta$.
$\delta^{(m)}$	m -dimensional Dirac delta distribution.
Δn^2	Dielectric or squared-index perturbation relative to the chosen optical reference.
$\delta \mathbf{E}$	Electric-field correction generated by scattering from Δn^2 .
δr_{DWBA}	Dimensionless first-order correction to the specular reflection amplitude.
$d\sigma / d\Omega$	Differential scattering cross-section per unit solid angle.
ϵ_0	Vacuum permittivity. It cancels between the Green-function normalization and the integral equation.
$\epsilon_i^\pm, \epsilon_f^\pm$	Local incident and reciprocal-exit polarization vectors for the two propagation directions.
$\mathbf{E}$	Total electric field in the actual medium.
$\mathbf{E}_0$	Exact electric field of the planar reference medium.
$f_j^0(\mathbf{Q})$	Normal atomic form factor of atom j .
f_j', f_j''	Real and imaginary anomalous dispersion corrections at the selected photon energy.
$\mathcal{F}, \mathcal{F}^{-1}$	Fourier and inverse Fourier transforms using the phase convention of Eq. (3).
$F_{\text{uc}}^{\text{atomic}}$	Atomic structure factor of one bulk unit cell before multiplication by the channel coefficient.
$F_{\text{atomic,DWBA}}$	Atomic DWBA matrix element; this is the quantity represented by the current bulk $\mathbf{F_DWBA}$.
$F_{\text{mean,DWBA}}$	DWBA matrix element of the homogeneous density already included in the optical reference.
$F_{\Delta, \text{DWBA}}$	Correct contrast matrix element, $F_{\text{atomic,DWBA}} - F_{\text{mean,DWBA}}$.
$\mathbf{G}_0$	Outgoing dyadic Green function of the planar reference medium, with normalization fixed by Eq. (20).
$\mathbf{H}$	Reciprocal lattice vector of the bulk crystal, used in the dynamical-theory Fourier expansion of Eq. (38).
K	Mean propagation constant of the crystal, $K^2 = k_0^2(1 + \chi_0)$, after absorption of χ_0 in Eq. (41).
L, l	Continuous rod coordinate $L = Q_z c / (2\pi)$ and the integer reciprocal lattice points l along the rod.
i	Imaginary unit, $i^2 = -1$.
$\mathbf{I}$	Identity tensor.
$\mathbf{k}_i, \mathbf{k}_f$	External incident and exit wavevectors.
$k_{z,j,i}, k_{z,j,f}$	Signed complex normal wavevectors of the incident and reciprocal exit fields in medium j .
κ_0	Positive vacuum normal-wavevector magnitude, $\kappa_0 = k_0 \sin \alpha = -k_{z,0}$.
κ_f	Positive exit-side vacuum normal-wavevector magnitude, $\kappa_f = k_0 \sin \alpha_f = k_0 \gamma_f$. Equal to κ_0 on the specular rod.

Symbol	Meaning
k_0	Vacuum wavevector magnitude, $2\pi/\lambda$, in inverse Angstrom.
λ	Vacuum X-ray wavelength, in Angstrom.
$\mathcal{L}_0$	Helmholtz operator of the unperturbed reference medium.
n, n_0	Actual and reference complex refractive indices.
$\nu = (\sigma_i, \sigma_f)$	One of the four Renaud incident/exit propagation channels.
o_j	Occupancy of atom j .
Ω	Solid angle around the selected exit direction.
ω	Angular frequency of the monochromatic X-ray field.
$\mathbf{p}$	Arbitrary source polarization or dipole vector acted on by the dyadic Green function.
$\mathbf{Q}$	External or channel-specific scattering transfer, according to context.
$\mathbf{Q}_{\parallel}$	Component of $\mathbf{Q}$ parallel to the surface.
Q_z	Channel-specific component of $\mathbf{Q}$ normal to the surface.
r	Distance from the coherent scattering region to the far-field observation point.
r_e	Classical electron radius, also called the Thomson radius.
r_0	Complex specular reflection amplitude of the prepared reference profile.
r_{total}	Coherent sum of reference and DWBA-correction reflection amplitudes.
R	Reflected intensity divided by incident intensity; for one polarization in vacuum, $R = r ^2$.
$\mathbf{r}, \mathbf{r}_{\parallel}, z$	Three-dimensional position and its components parallel and normal to the surface.
$\mathbf{r}_j$	Position of atom j .
ρ_a	Atomic number density for a homogeneous elemental or effective medium.
$\rho_f(\mathbf{r})$	Complex atomic scattering-factor density in electron units per volume.
$\bar{\rho}_f$	Homogeneous cell-average complex scattering-factor density used by the bulk optical reference.
σ_i, σ_f	Channel direction signs, each equal to +1 or -1.
$\Theta(-z)$	Indicator function of the substrate half-space.
$\chi, \chi_{\mathbf{H}}, \chi_0$	Susceptibility $\chi = n^2 - 1$, its Fourier coefficients over $\mathbf{H}$, and the mean coefficient $\chi_0 = -2(\delta_0 + i\beta_0)$.
V_{uc}	Bulk unit-cell volume.
$W_j(Q^2)$	Debye–Waller exponent of atom j , possibly evaluated at complex Q^2 .
z_j^{ref}	Local z position at which stored layer amplitudes are referenced.

12 References

The Green-function and reciprocal-field construction follows G. Renaud, R. Lazzari, and F. Leroy, “Probing surface and interface morphology with grazing incidence small angle X-ray scattering,” *Surface Science Reports* **64** (2009), 255–380, especially Sections 5.2–5.4. The notation in this design note has been adapted to the signed k_z , $A^\pm$, electric-field, and uppercase- Q conventions used by orGUI.

The works below were used as external cross-checks; the companion note `dwba_literature_crosscheck.md` records the comparison in detail. No published treatment of the mean-density subtraction on the *specular* rod was found. The DWBA crystal-truncation-rod literature works on non-specular rods, where Eq. (50) makes the term vanish, and the specular reflectivity literature

works with continuum density profiles, where the crystalline lattice sum never appears. The combination therefore has to be validated numerically rather than by citation.

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